

1 Problem Setup

1D inviscid Burgers Riemann problem

Governing equation

$$u_t + u u_x = 0, \quad x \in [-1, 1], \quad t \in (0, 1]$$

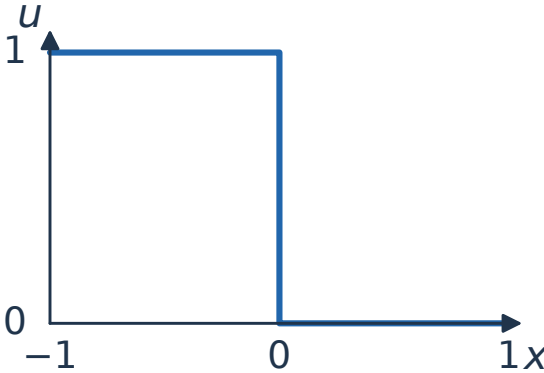
Initial condition

$$u(x, 0) = 1 \quad (x < 0), \quad u(x, 0) = 0 \quad (x > 0)$$

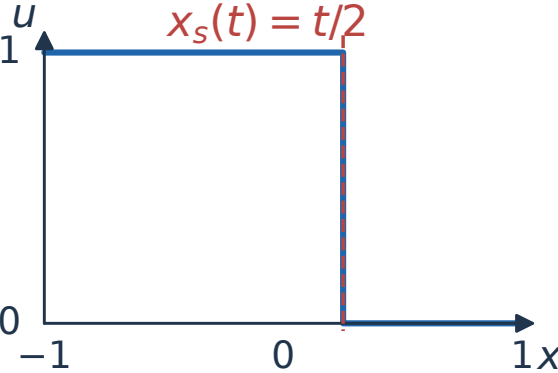
Boundary conditions

$$u(-1, t) = 1, \quad u(1, t) = 0$$

Initial condition



Entropy shock: t = 0.5



$x_s(t) = t/2, \quad s = (u_L + u_R)/2 = 0.5$
The faster left state forms a moving shock.
Blue profiles: analytical inviscid reference.

2 PINN Framework

Coordinate network (schematic)

